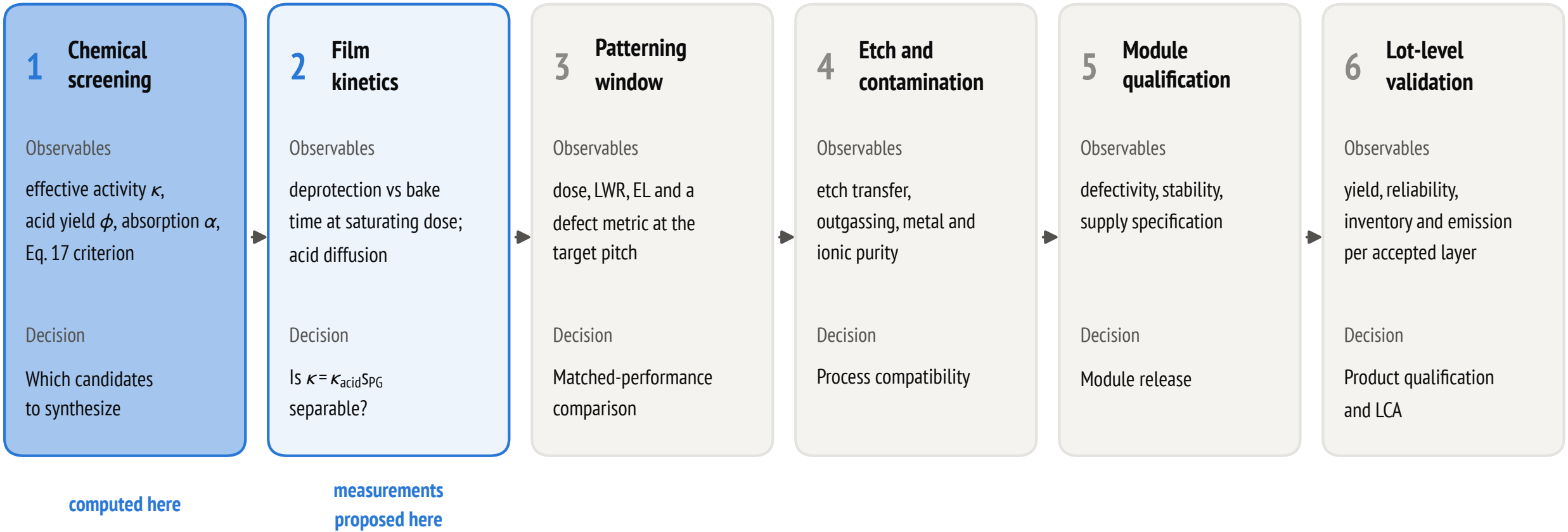
1
Chemical screening
Observables
effective activity κ, acid yield ϕ, absorption α, Eq. 17 criterion
Decision
Which candidates to synthesize
2
Film kinetics
Observables
deprotection vs bake time at saturating dose; acid diffusion
Decision
Is $\kappa = \kappa_{acid} S_{PG}$ separable?
3
Patterning window
Observables
dose, LWR, EL and a defect metric at the target pitch
Decision
Matched-performance comparison
4
Etch and contamination
Observables
etch transfer, outgassing, metal and ionic purity
Decision
Process compatibility
5
Module qualification
Observables
defectivity, stability, supply specification
Decision
Module release
6
Lot-level validation
Observables
yield, reliability, inventory and emission per accepted layer
Decision
Product qualification and LCA
computed here
measurements proposed here